

why React?

1. Component - Based Architecture:

React uses a component based architecture where the UI is divided into small, reusable and independent components, making application easier to develop, maintain and scale.

2. Reusable components:

React allows us to create components once and reuse them in different parts of the application.

Eg: Button, Number, Card.

3. Declarative UI:

We describe what the UI should look like, and React handles how to update it.

This makes code easier to understand and maintain.

4. Virtual DOM:

React uses Virtual DOM to efficiently determine what part of the UI needs to change instead of updating everything.

5. SPA - (Single Page Application)

* A SPA is a web Application where the browser loads one main HTML Page, and React change the content on that page without doing a full page reload for every navigation.

* Simple Answer: SPA $\rightarrow$ Single HTML Page + Dynamic content + No Full Page reload.

6. One-way data flow:

* one way data flow in React means data flow from parent component to child components through props.

* This make the Application easier to understand, debug and maintain. Data flows downward.

7. Diffing algorithm:

* The diffing algorithm in React process of comparing the previous virtual Dom with the updated virtual Dom to identify the difference and update only the necessary part of the real Dom.

8. Reconciliation

* Reconciliation is React's process of comparing the previous and new virtual DOM trees after a state or props change and efficiently updating the required parts of the Real DOM.

Difference Reconciliation vs Diffing

Reconciliation	Diffing
* Overall process of determining how the UI should be updated	* comparing the old and new UI trees to find difference.
* Decide what needs to happen	* Finds what changed.
* Leads to the required DOM update	* Identifies the difference.

9. Dynamic Rendering:

* Dynamic re-rendering in React is the process of automatically update a component's UI when its state or props change.

* React determines the required change and updates the DOM efficiently.